Installing & Setting Up Git and GitHub in Visual Studio

1 Comment / By Dot Net Tutorials / February 22, 2026

Back to: Microservices using ASP.NET Core Web API Tutorials

Installing & Setting Up Git and GitHub in Visual Studio

Modern software development is never about writing code alone. Real-world projects involve **continuous changes, collaboration, backups, history tracking, and teamwork**. This is where **Git** and **Visual Studio's Git integration** become essential.

Git is a **Version Control Tool** that helps us track changes to our code over time, revert to older versions, and work safely without fear of breaking everything. Visual Studio has **built-in Git integration**, so we can do most Git tasks (commit, push, pull, branch, merge) without leaving the IDE.

What is Git?

Git is a **free and open-source Distributed Version Control System (DVCS)** used to:

- Track changes made to source code files
- Maintain a complete history of a project
- Allow multiple developers to work safely on the same codebase
- Restore older versions when something goes wrong

Git treats your code as a **timeline of snapshots**, not just files. It is a **tool/software**, not a website, and not part of Visual Studio by default.

Do We Need to Install Git Separately for Visual Studio?

Yes. 100% yes. This is one of the most misunderstood points for beginners.

- Visual Studio **does NOT ship with Git.**
- Visual Studio only provides a **graphical interface.**
- Git is the **Actual Engine** that performs version control.

Think of it like this:

- Git = Engine
- Visual Studio = Dashboard & controls

Without Git installed:

- Git menu options in Visual Studio won't work correctly

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- You won’t be able to create or manage repositories

So, the **first and mandatory step** is installing Git on Windows.

How to Install Git on Windows

To use Git with Visual Studio 2022, Git must first be installed on your Windows system. Git for Windows provides all the tools required to track source code changes and interact with remote repositories such as GitHub. The following steps explain **how to download, install, and configure Git on Windows** using recommended settings suitable for beginners and Visual Studio users.

Step 1: Download Git for Windows

1. Open your browser
2. Navigate to the official Git website: **https://git-scm.com**
3. Click **Download for Windows**
4. The latest **Git for Windows installer (.exe)** will download automatically

For a better understanding, please have a look at the image below.



Choosing the Correct Installer

On the download page, you may see multiple options. For a **typical 64-bit Windows PC (Intel/AMD)**, choose **Git for Windows / x64 Setup (Standalone Installer)**. This is the **standard and recommended installer**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



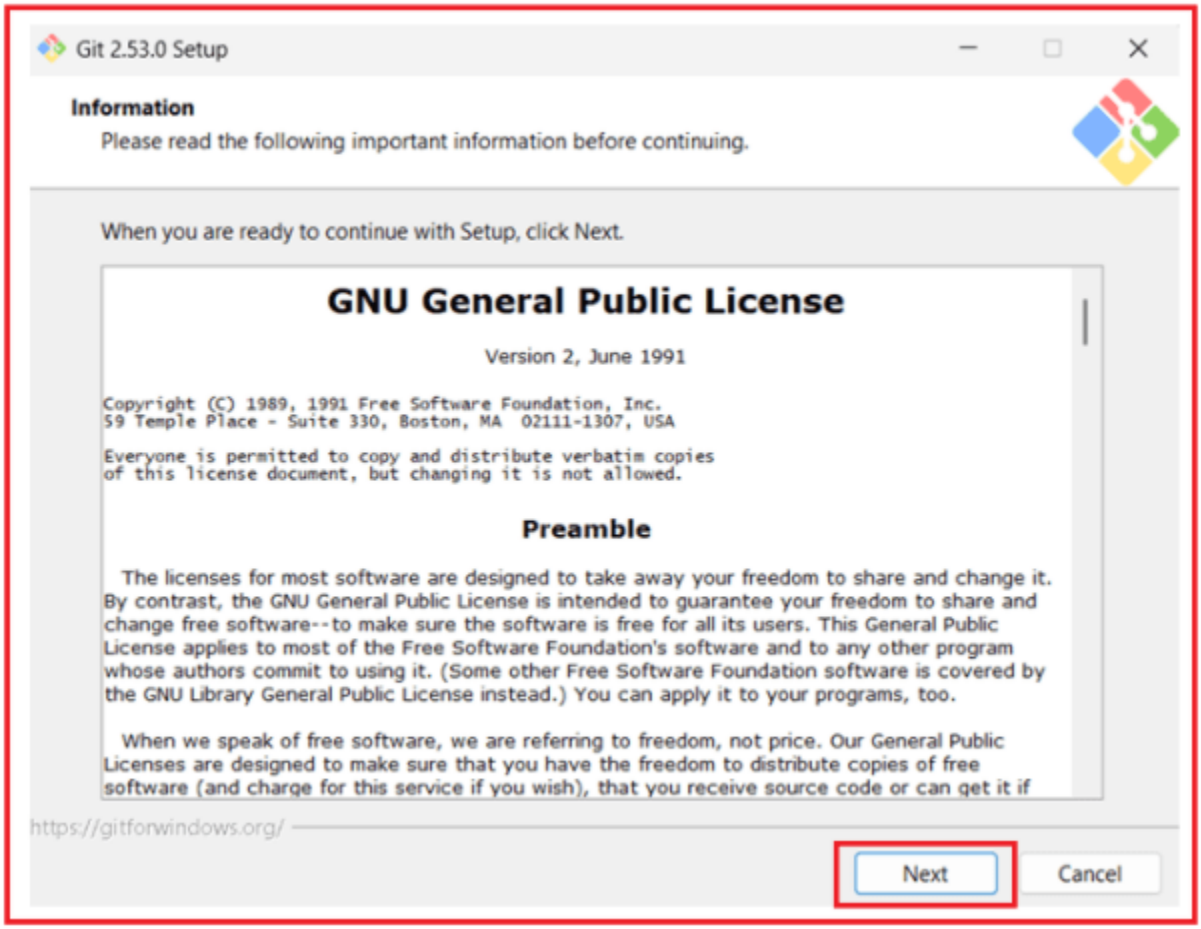
This installer is best for:

- Windows 10 / Windows 11
- Visual Studio 2022 users
- Beginners
- Regular desktops and laptops (Intel / AMD CPUs)

Step 2: Start the Git Installer

1. Double-click the downloaded .exe file
2. The Git Setup Wizard opens
3. Click **Next** on the welcome screen

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Step 3: Important Installation Screens

Most beginners get confused here. Follow these **recommended settings**.

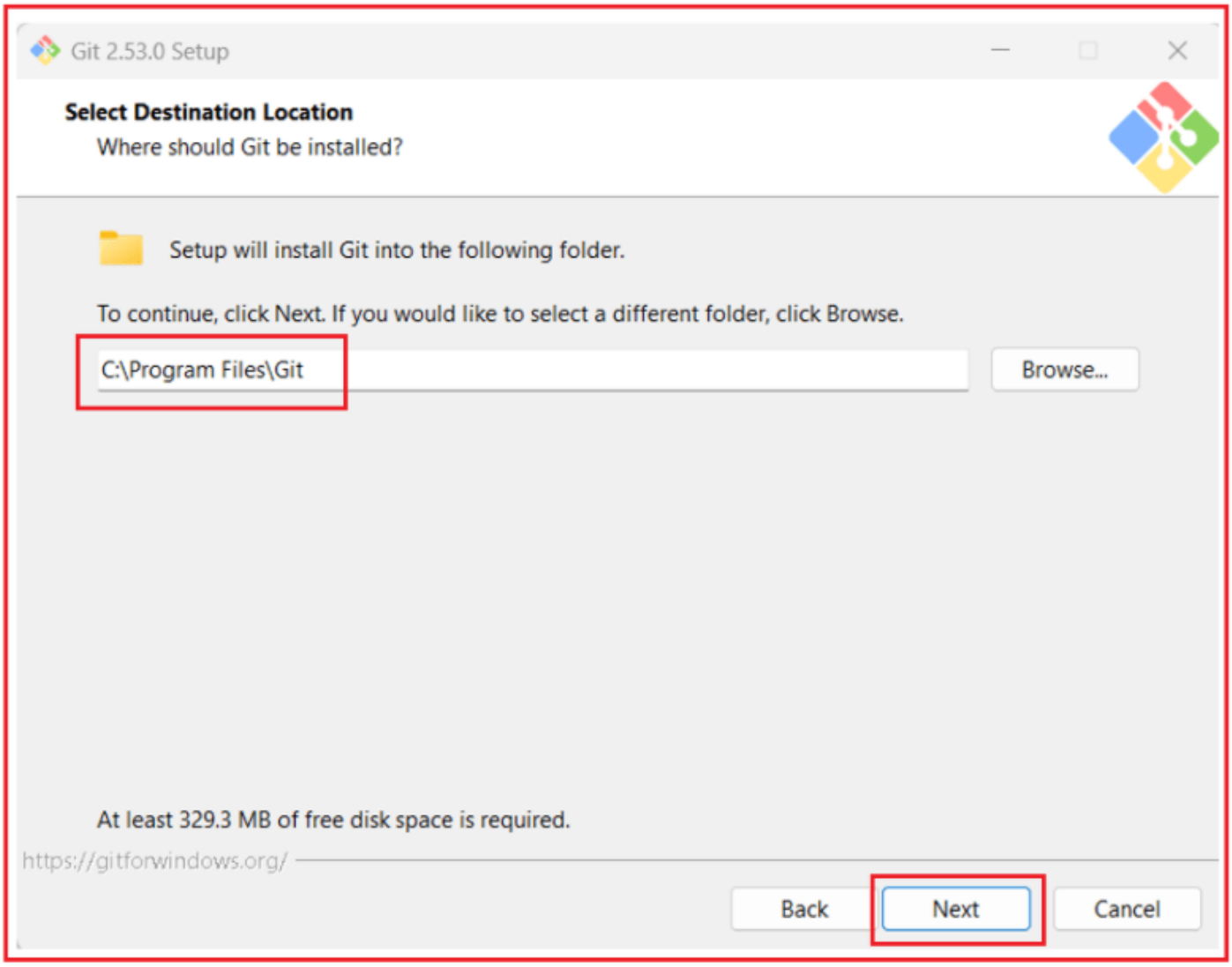
Selecting the Destination Folder

Keep the default location:

- C:\Program Files\Git
- Click **Next**

The default path is stable, widely used, and works perfectly with Visual Studio.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Selecting Components

These options decide what extra features Git installs.

Keep the following options checked:

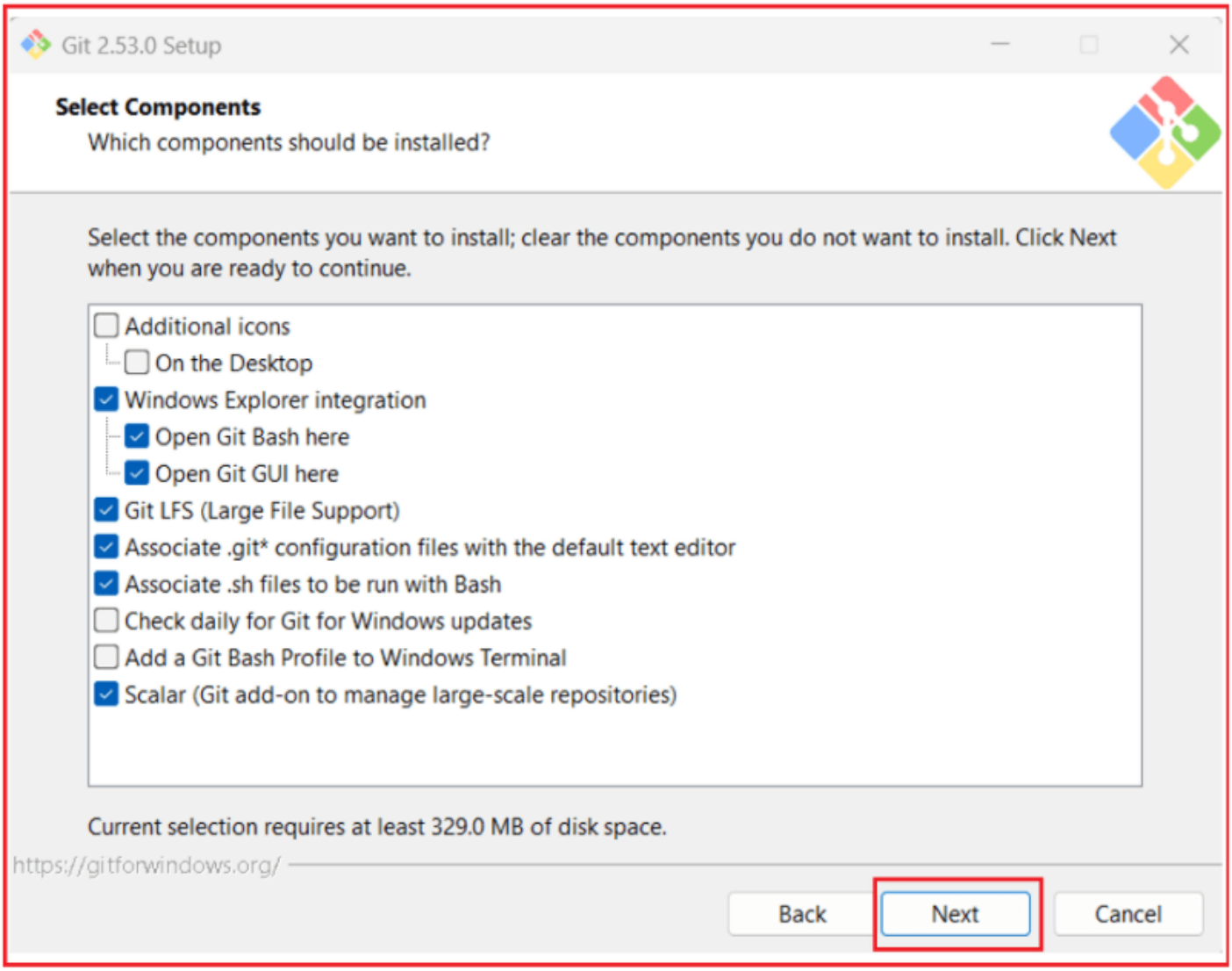
- **Windows Explorer integration**
  - Open Git Bash here
  - Open Git GUI here
- **Git LFS (Large File Support)**
- **Associate .git*** configuration files with the default editor
- **Associate .sh files to be run with Bash**
- **Scalar (Git add-on to manage large-scale repositories)**

Leave the following options unchecked:

- Additional desktop icons
- Check daily for Git for Windows updates
- Add a Git Bash profile to Windows Terminal

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Click **Next** to continue. These settings provide all essential Git features without cluttering your system.

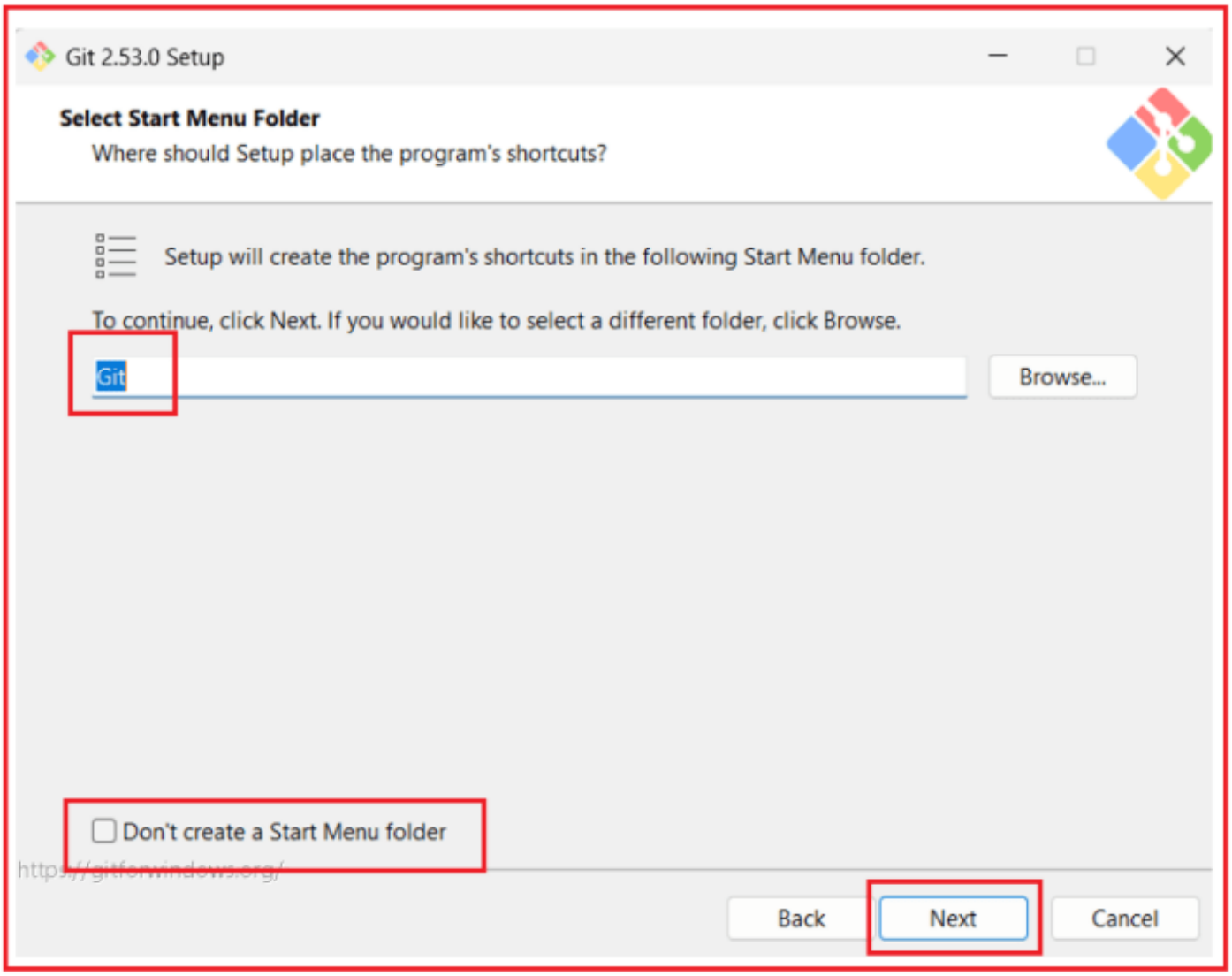


Select Start Menu Folder

This only controls where Git shortcuts appear.

- Keep the folder name as **Git** (default)
- Do **not** check “Don’t create a Start Menu folder.”
- Click **Next**

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Selecting the Default Editor

Git sometimes opens a text editor for:

- Commit messages
- Configuration files

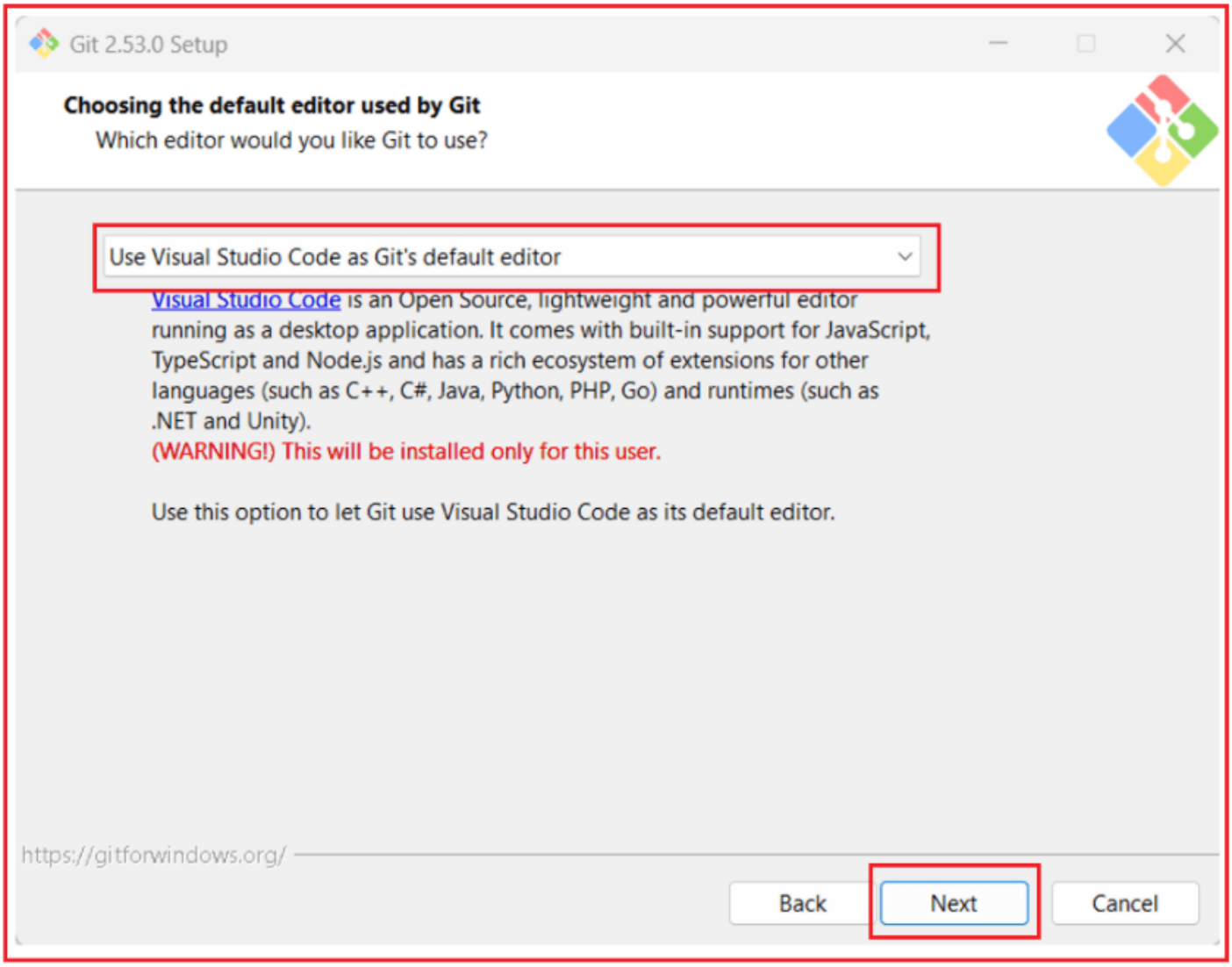
Select: **Use Visual Studio Code as Git's default editor**

Why this choice?

- Clean and modern UI
- Beginner-friendly
- Industry standard
- Easy to understand commit messages

Click **Next**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Note: The warning “This will be installed only for this user” is normal and safe. It simply means VS Code will be available for your Windows User Account.

Adjusting the Name of the Initial Branch

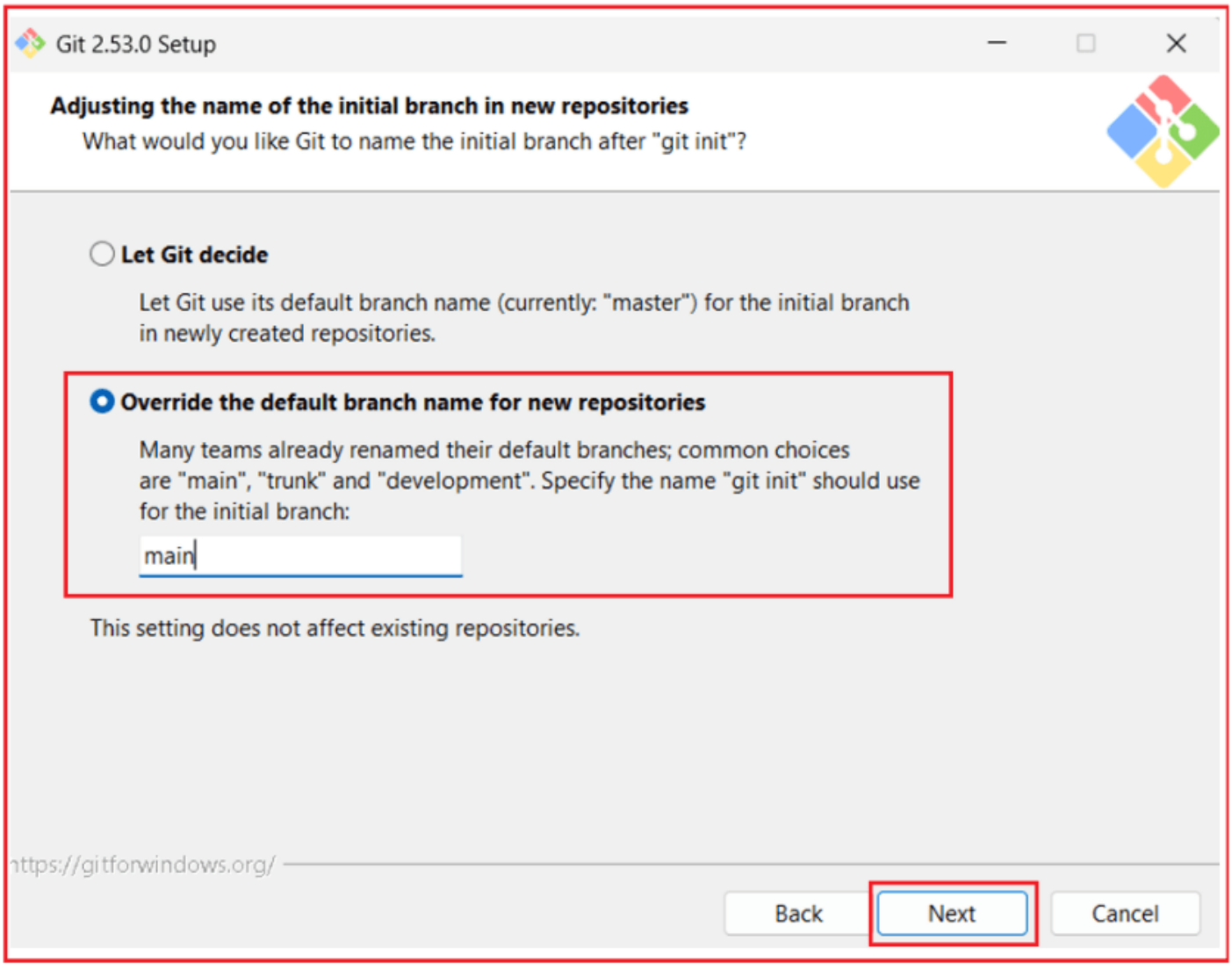
This decides the default branch name when creating new repositories.

Select:

- **Override the default branch name**
- Set the branch name to: **main**

Click **Next**. This follows modern Git and GitHub standards and avoids confusion later.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Adjusting PATH Environment

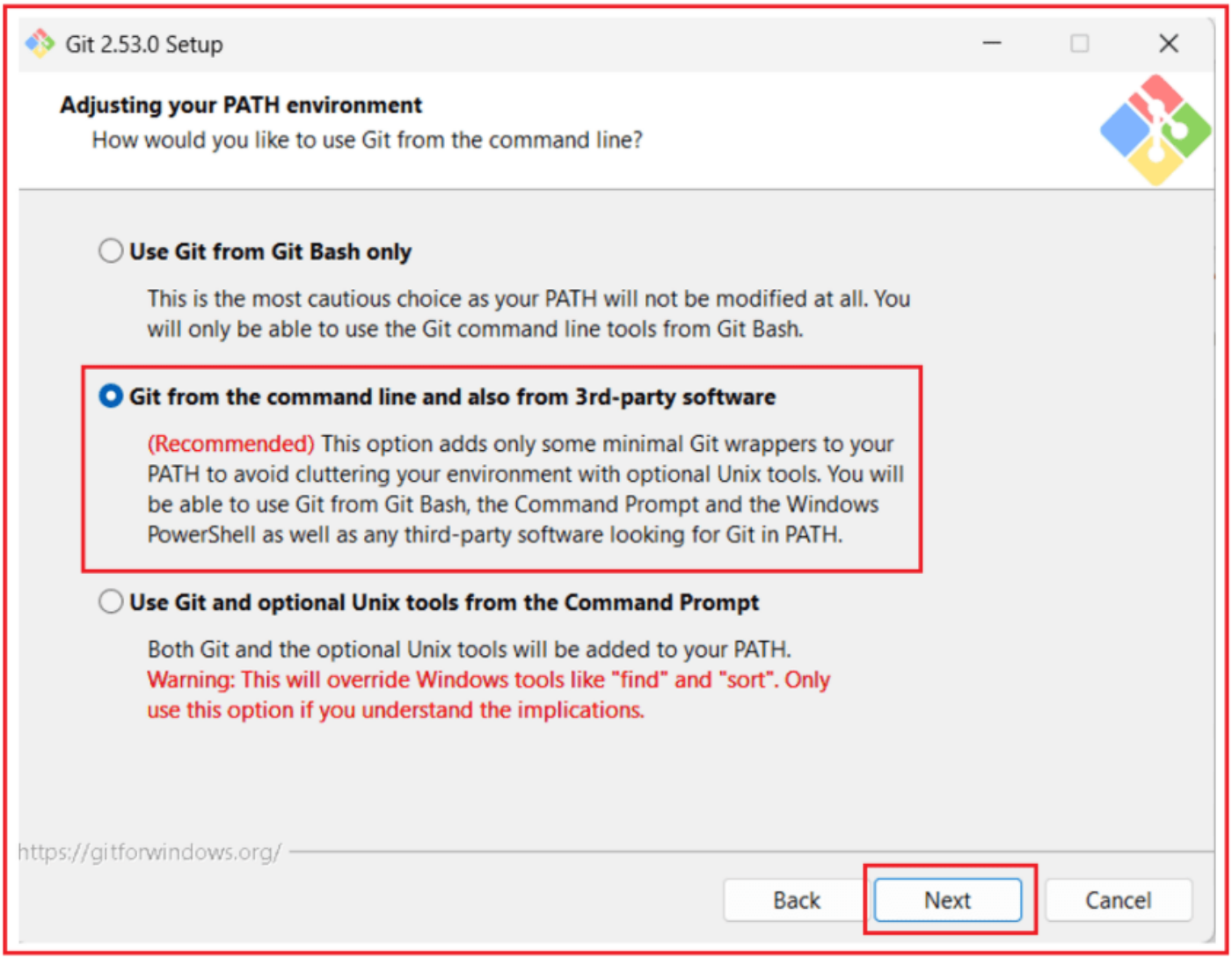
Select: **Git from the command line and also from 3rd-party software**

This option:

- Allows Git to work with:
  - Git Bash
  - Command Prompt
  - PowerShell
  - Visual Studio
- Does not override Windows system tools

Click **Next**. This is the **recommended and safest choice**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Choosing the SSH Executable

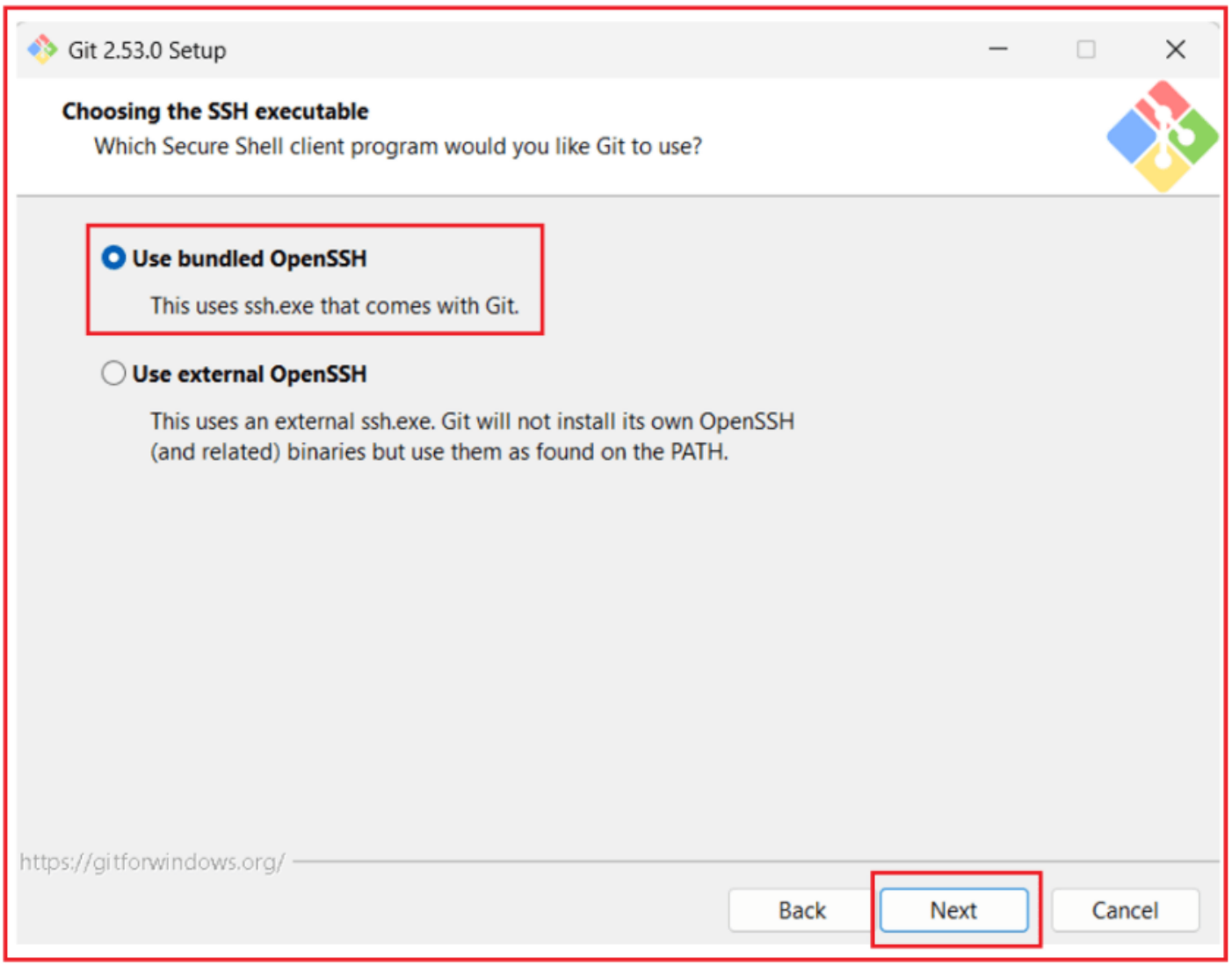
Select: **Use bundled OpenSSH**

This allows Git to use the SSH client bundled with Git.

- Click **Next**

No extra configuration required.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Choosing HTTPS Transport Backend

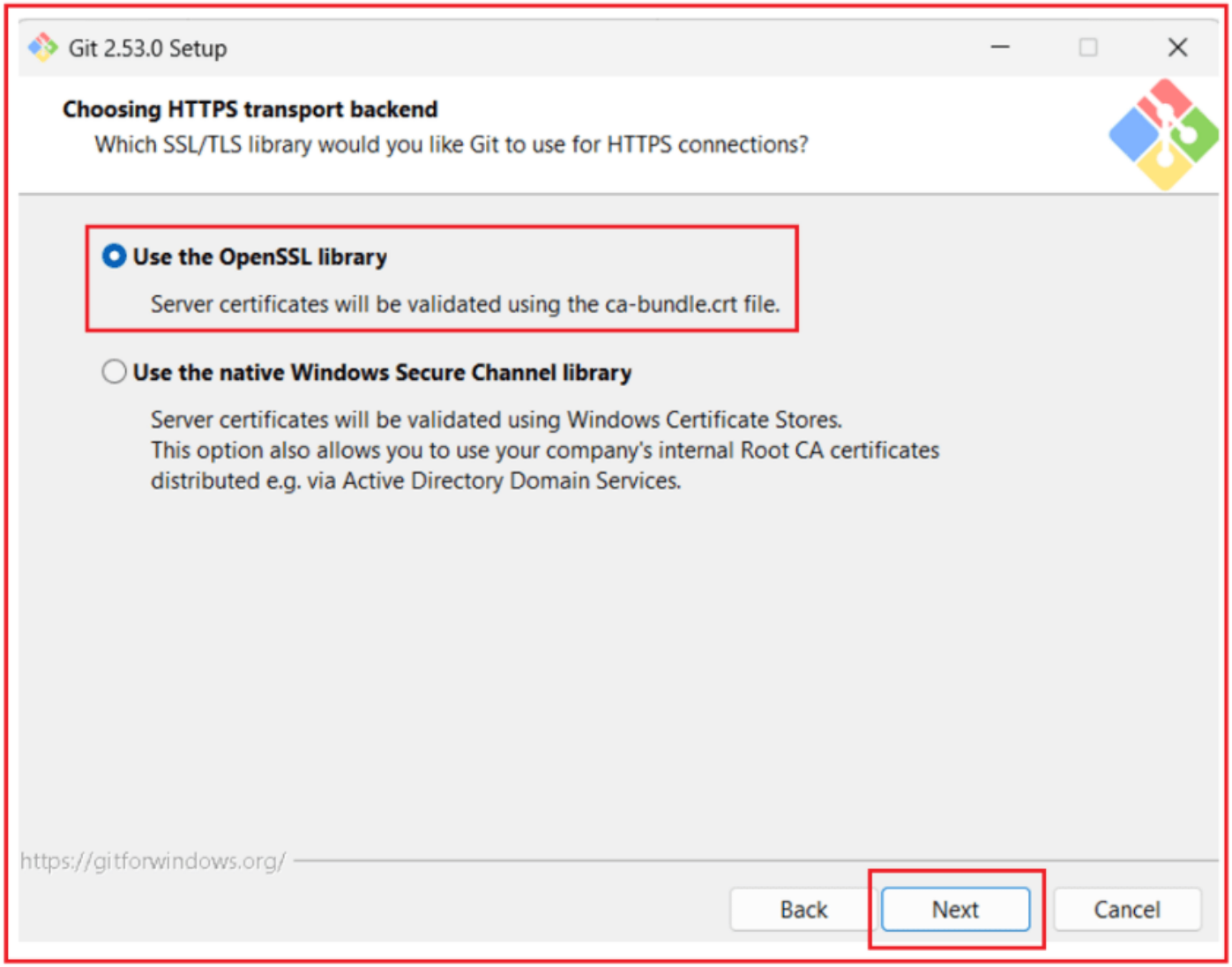
Select: **Use the OpenSSL library**

This allows

- Secure HTTPS communication with GitHub and other services.

Click **Next**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Configuring Line Ending Conversions

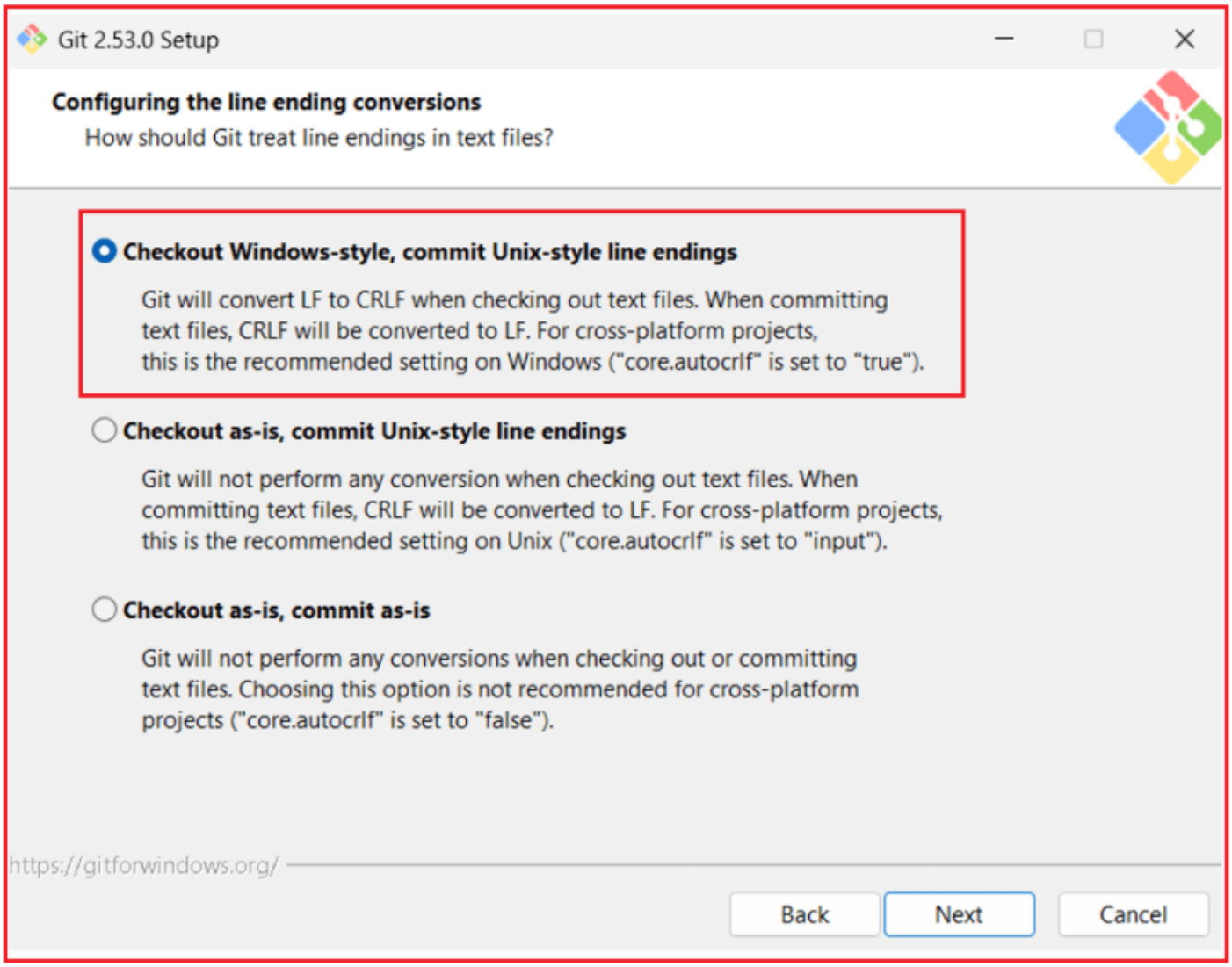
Select: **Checkout Windows-style, commit Unix-style line endings**

This ensures:

- Compatibility across Windows, Linux, and macOS
- Fewer line-ending issues in team projects

Click **Next**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Configuring the Terminal Emulator for Git Bash

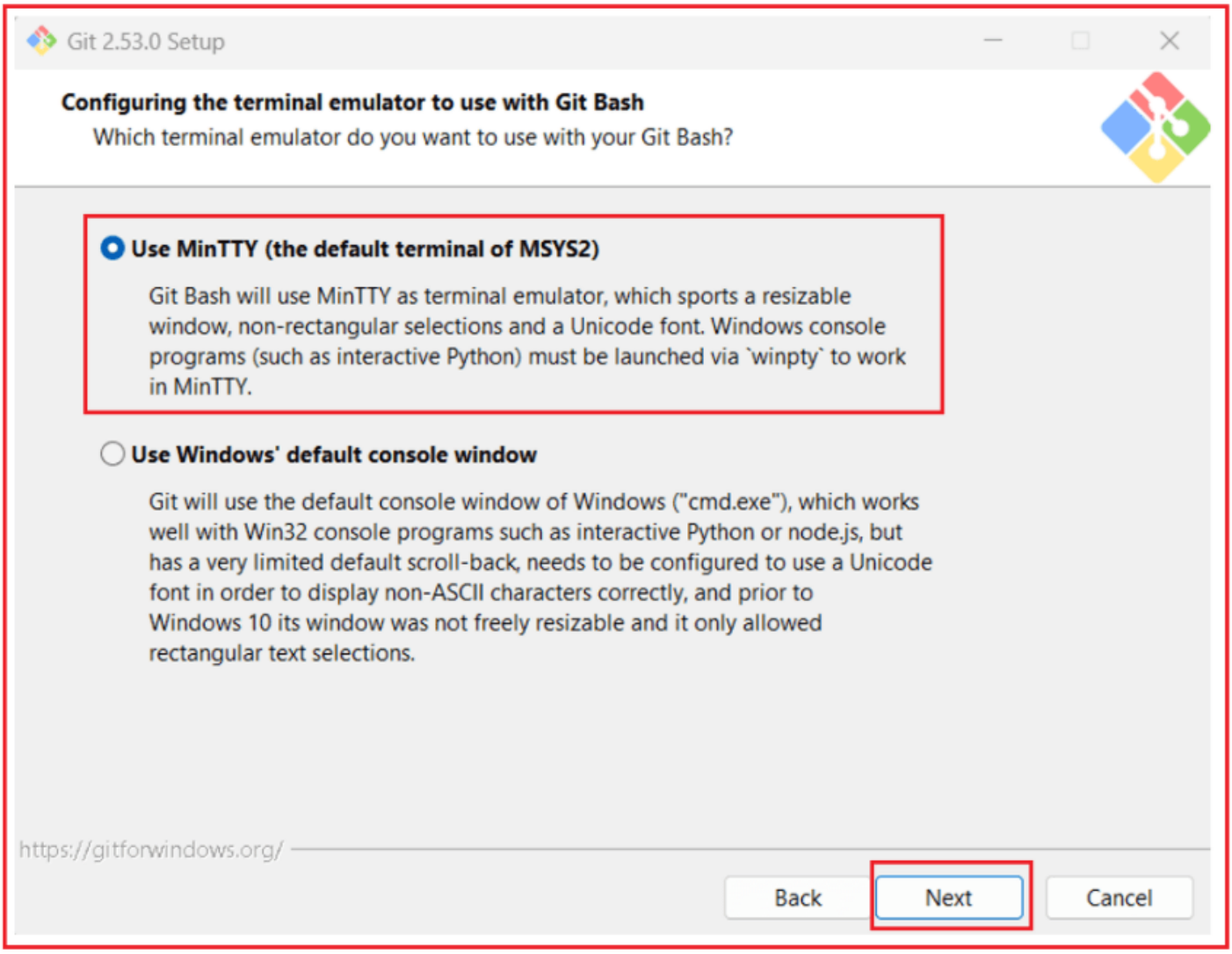
Select: **Use MinTTY (the default terminal of MSYS2)**

Benefits:

- Resizable terminal window
- Unicode support
- Modern command-line experience

Click **Next.**

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Choose the Default Behaviour of git pull

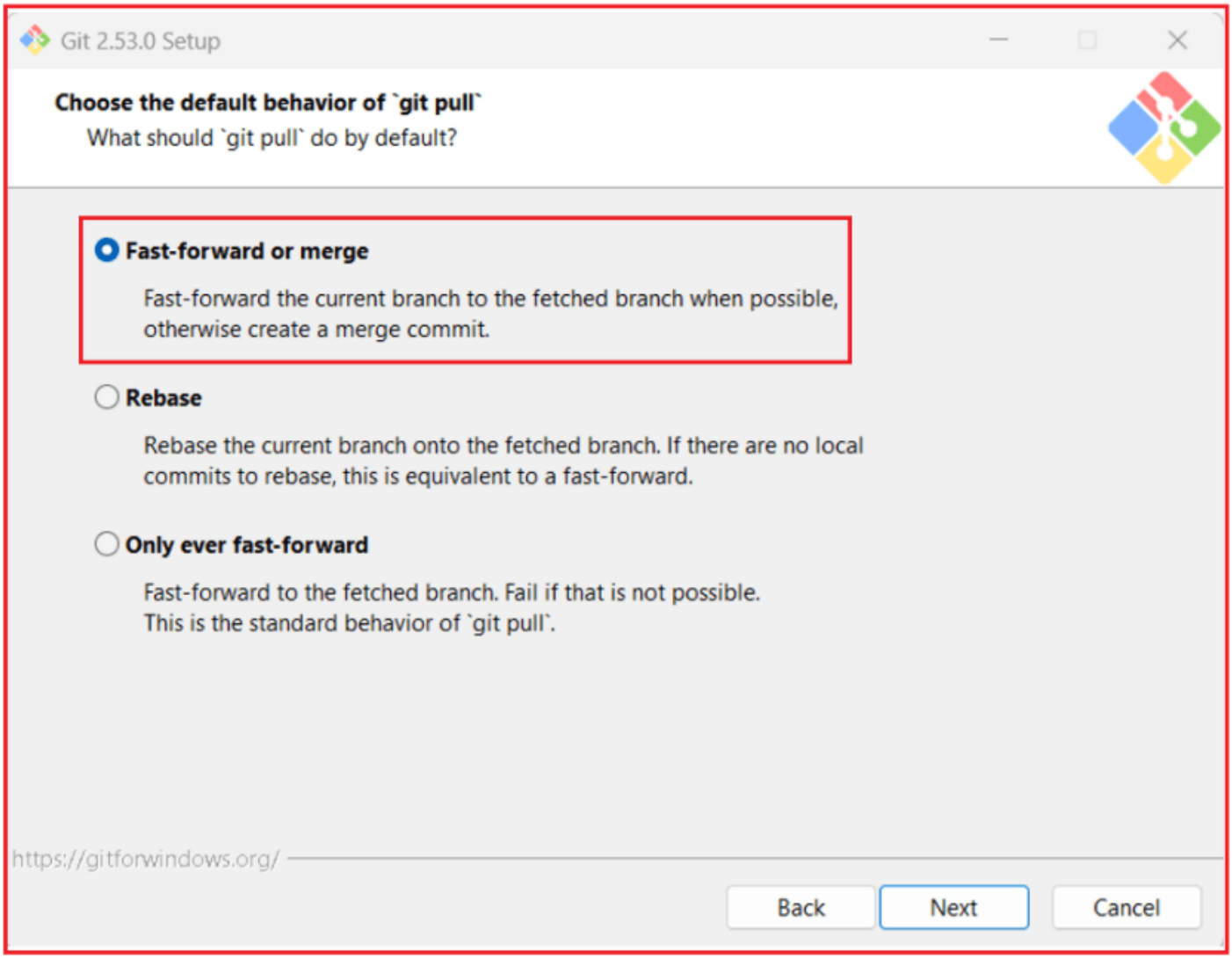
Select: **Fast-forward or merge**

This option:

- Keeps history simple
- Works well for beginners
- Avoids unexpected failures

Click **Next.**

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



Choose a Credential Helper

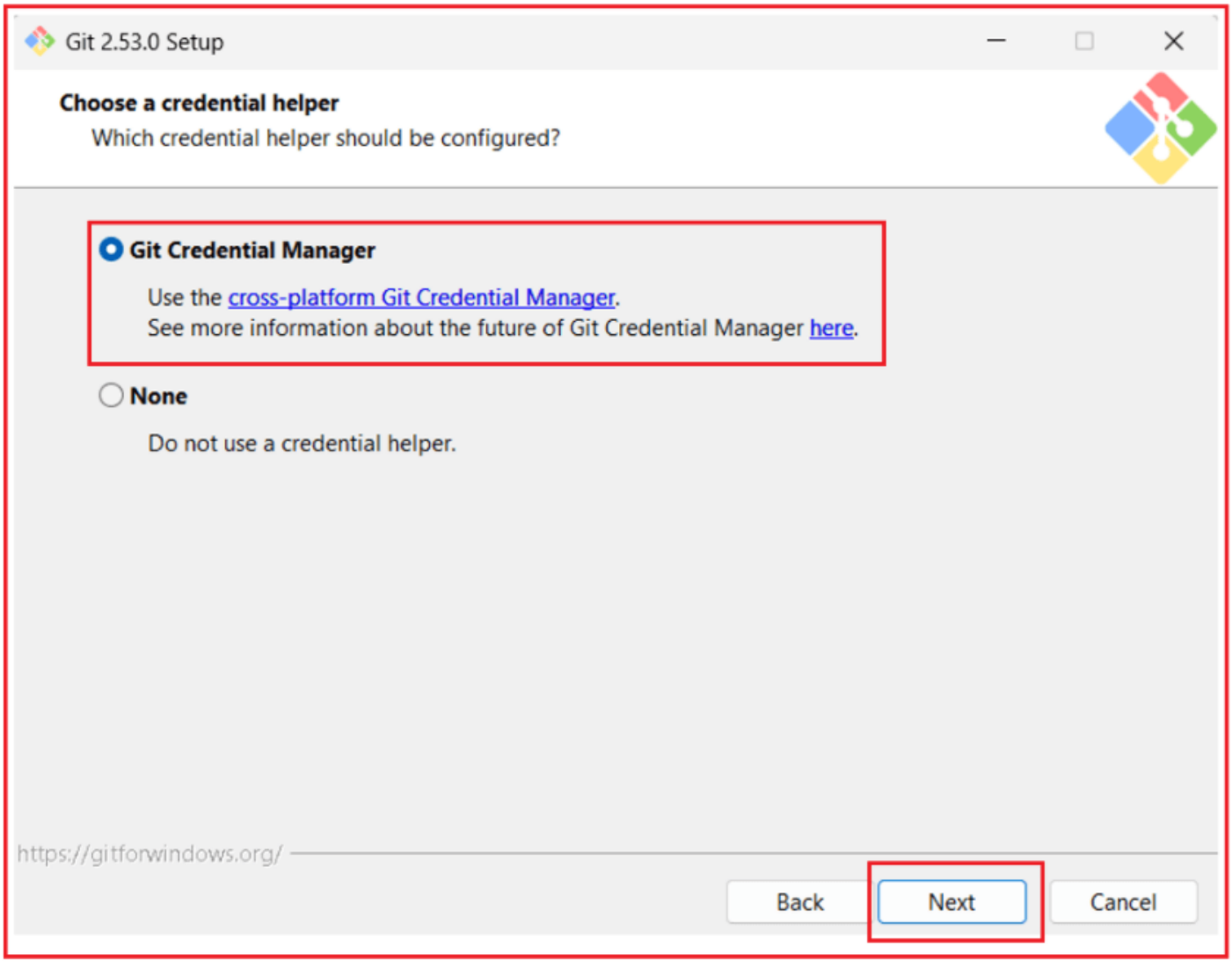
Select: **Git Credential Manager**

This allows Git to:

- Securely store authentication tokens
- Avoid repeated login prompts
- Integrate smoothly with GitHub

Click **Next**.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



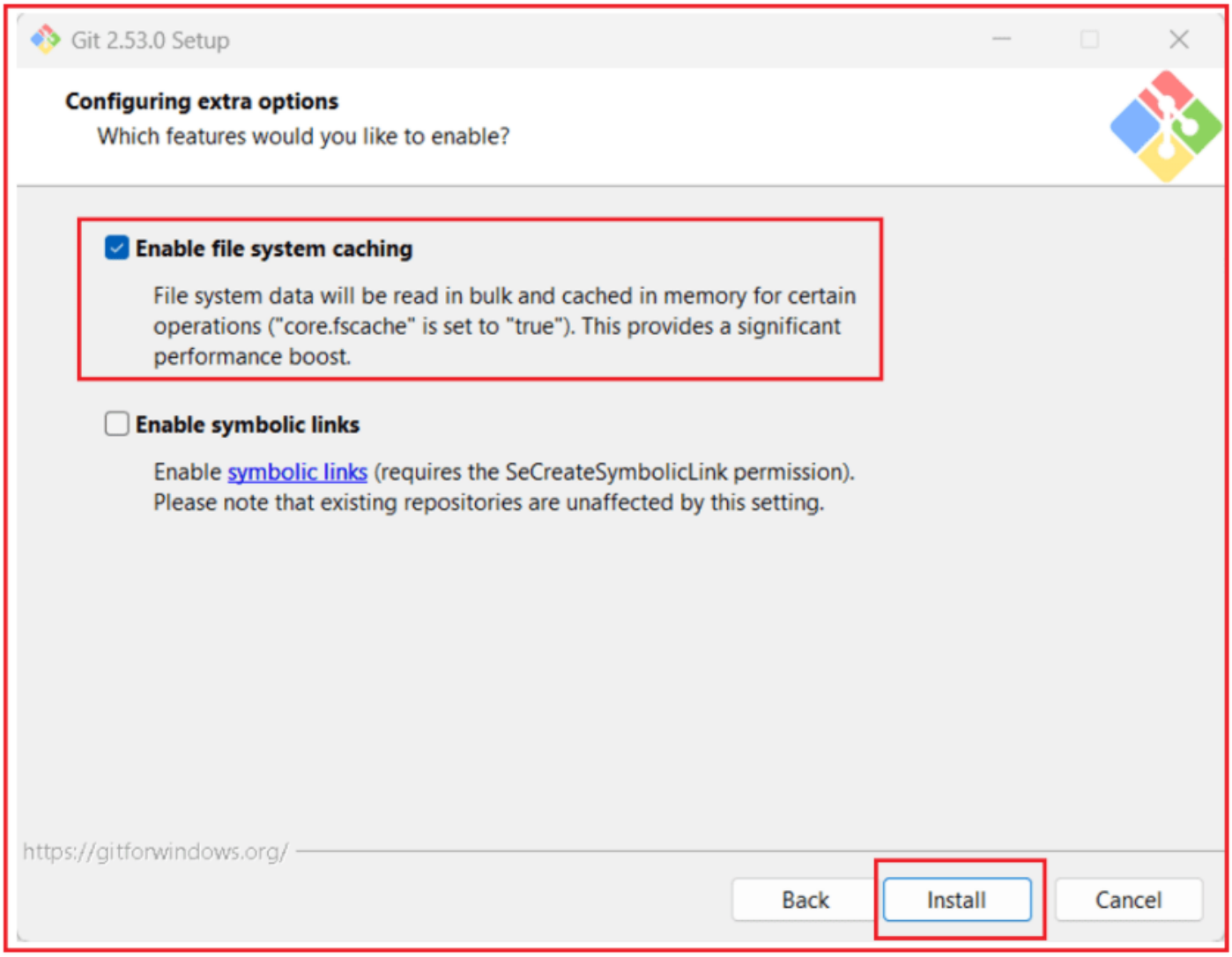
Configuring Extra Options

This screen allows you to enable **additional performance and system-related features** for Git.

- Keep **Enable file system caching** selected to improve Git performance.
- Leave **Enable symbolic links** unchecked unless your project specifically requires symbolic links.

Click **Install** to complete the setup.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



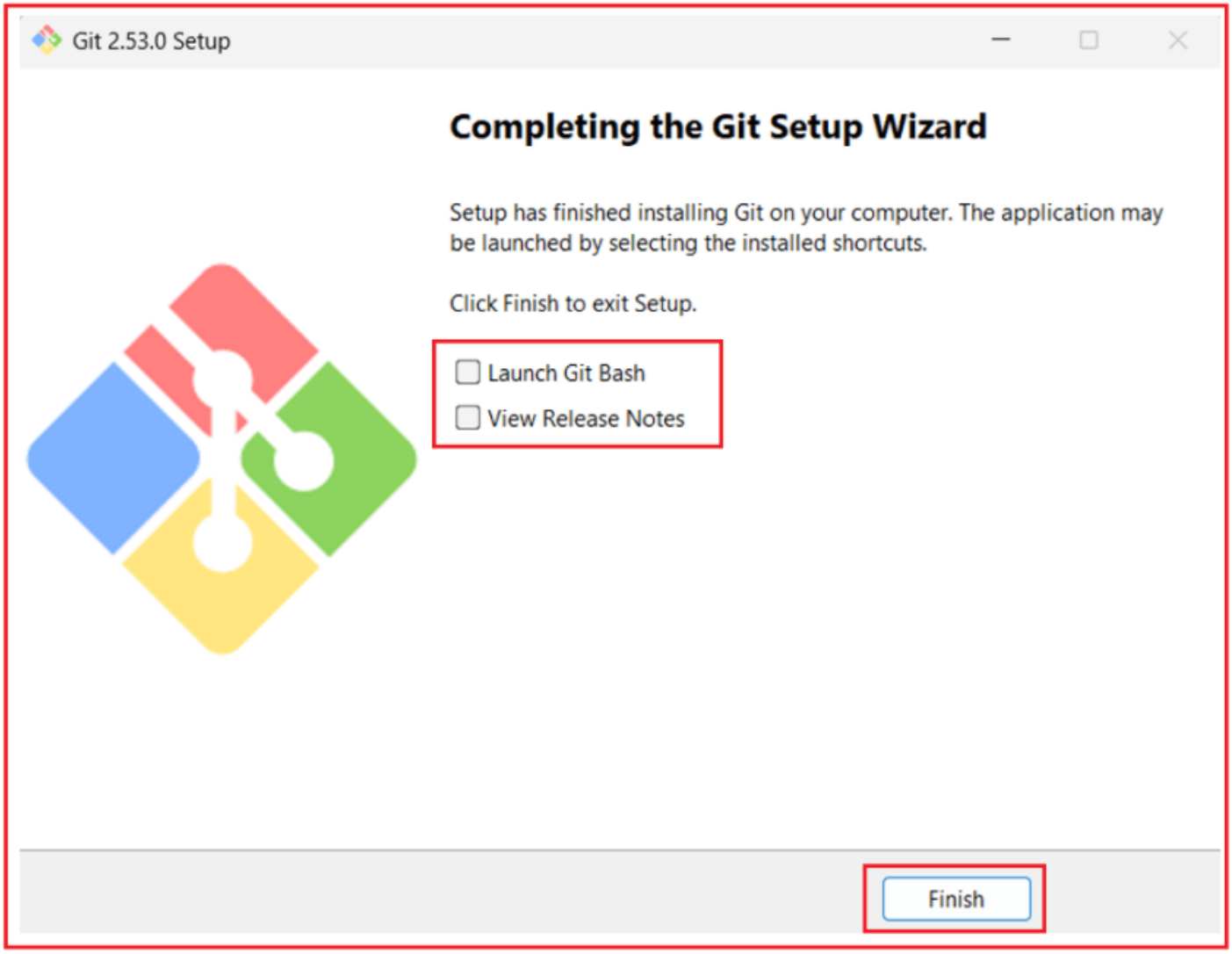
Completing the Git Setup Wizard

Once installation completes:

- Uncheck **Launch Git Bash**
- Uncheck **View Release Notes**
- Click **Finish**

Git is now successfully installed and ready to be used with Visual Studio.

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF



How to Verify Git Installation?

After installing Git, verification is an important step. It ensures that Git is correctly installed on your system and available to tools such as Visual Studio. Verification can be done in multiple ways, and each method confirms a different aspect of the installation.

Method 1: Using Command Prompt

This method verifies that Git is installed at the operating system level and available in the system PATH.

Steps:

- Open **Command Prompt**
- Type the following command and press Enter:
  - **git –version**

What this confirms:

- Windows can locate the Git executable

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- Git is correctly installed
- PATH environment variables are configured properly

Expected Output:

```
C:\Users\Pranaya>git --version
git version 2.53.0.windows.1
```

If you see a version number, Git is installed correctly. If you see an error such as "**git is not recognized as an internal or external command**", it means:

- Git is not installed correctly, **or**
- Git is installed, but not added to PATH

Method 2: Using Visual Studio 2022

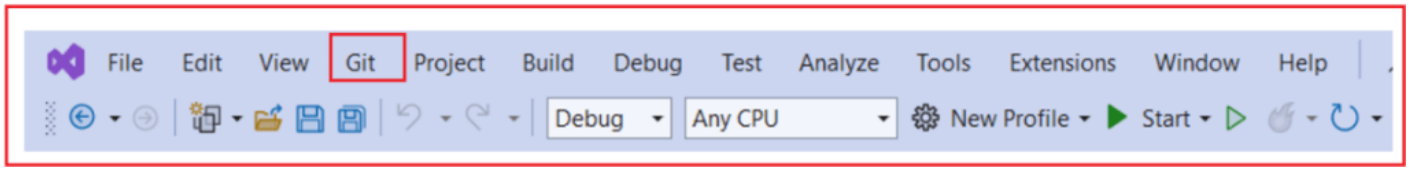
This method verifies that Git is not only installed, but also **integrated with Visual Studio**.

Steps:

1. Open **Visual Studio 2022**
2. Open any:
    - Project
    - Solution
    - Or local folder
3. Once the project loads, observe the **top menu bar**

What to look for:

- The **Git menu** appears between **View** and **Project**



What this confirms:

- Git is installed on the machine
- Visual Studio has detected Git
- Visual Studio can communicate with Git successfully

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

Seeing the Git menu confirms that Git is **installed, detected, and ready to use inside Visual Studio**.

How does Visual Studio integrate with Git?

Visual Studio is **not Git itself**. Git is a separate version control tool installed on your system. Visual Studio serves as a **graphical user interface (UI)** that communicates with Git on your behalf. When you perform Git-related actions in Visual Studio, you are not bypassing Git—you are **using Git indirectly**.

Behind the scenes:

- Visual Studio sends instructions to Git
- Git performs the actual operation
- Visual Studio displays the results visually

For example:

- Clicking **Commit** → runs a Git commit
- Clicking **Push** → runs a Git push
- Clicking **Create Branch** → runs a Git branch command

This allows beginners to use Git effectively **without memorizing commands**, while still using Git's full power.

Why Git Alone Is NOT Enough for Sharing?

Git works **locally**. This means it tracks changes only on your computer.

With only Git installed:

- Your repository exists only on your system
- Other people cannot access your code
- There is no online backup

Git is excellent for:

- Tracking changes
- Managing history
- Working with branches

But Git **does not provide any online sharing mechanism** by itself. To share code with students or teams, a **remote hosting platform** is required.

What is GitHub?

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

GitHub is a **Cloud-Based Platform** that hosts Git repositories and makes them accessible over the internet. It allows developers and teams to store, share, and collaborate on Git-based projects.

In simple terms, **GitHub is the online home for Git repositories.** GitHub does not replace Git. Instead, GitHub **works with Git** by providing:

- Remote Storage
- Sharing
- Collaboration tools

GitHub can be accessed from anywhere using a web browser. Official website: **https://github.com**. GitHub is **not installed** on your computer.

Why GitHub is Used:

- Store repositories online
- Share code with students or teams
- Maintain online backups
- Collaborate with others
- Use features like issues, pull requests, CI/CD

Difference Between Git and GitHub

This is one of the most important concepts for beginners. Git manages code changes locally, while GitHub hosts Git repositories online for sharing and collaboration.

Git:

- A version control system
- Installed on your computer
- Tracks changes and history
- Works offline
- Manages branches and commits

GitHub:

- A hosting service
- Runs on the internet
- Stores Git repositories online
- Enables sharing and collaboration
- Requires an account

Relationship:

- Git can work without GitHub

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- GitHub cannot work without Git

How to Register with GitHub?

A GitHub account is required to host and share repositories online and can be created for free.

Step-by-step registration

1. Go to **https://github.com**
2. Click **Sign up**
3. Enter:
   - Email address
   - Password
   - Username
   - Country/Region
4. Verify your email
5. Complete basic setup questions

That's it. You now have a **GitHub account**.

How to Sign in to GitHub from Visual Studio?

Once Git and GitHub are ready, you can connect Visual Studio to your GitHub account.

Step 1: Open Visual Studio

Go to:

File → Account Settings

Step 2: Sign In

- Click **Add Account**
- Choose **GitHub**

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

This will open your browser, and you'll need to log in with your GitHub account details. Then, it will open the following **GitHub Authorization** page.

GitHub Authorization Page

This screen appears when Visual Studio requests permission to access GitHub on your behalf. It allows Visual Studio to perform Git operations without repeatedly asking for credentials. This process is:

- Secure
- Official
- Required only once

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

You are **not sharing your password** with Visual Studio.

What Is Actually Happening Behind the Scenes?

When you clicked **Sign in to GitHub from Visual Studio**:

- Visual Studio redirected you to GitHub
- GitHub is about to create a **Personal Access Token (PAT)** automatically
- This token will be stored securely and used by Visual Studio
- GitHub is showing **exactly what permissions** Visual Studio will have

You are **not sharing your password**.

What is a Personal Access Token (PAT)?

A **Personal Access Token (PAT)** is a **secure authentication token** used in place of a password.

It looks like: ghp_xxxxxxxxxxxxxxxxxxxx

PATs:

- Are generated from GitHub
- Have limited permissions

Explore our developer-friendly HTML to PDF API Printed using PDFCrowd HTML to PDF

- Can be revoked anytime

Why PAT Is Used Instead of Passwords?

GitHub **no longer allows password authentication** for Git operations.

Reasons:

- Passwords are insecure
- Tokens can be limited and revoked
- Better protection against attacks

Visual Studio:

- Automatically uses PAT after sign-in
- Stores it securely using Windows Credential Manager

You don’t usually need to manually create PAT when using the Visual Studio login.

Conclusion:

Now that Git is installed and verified on your system, Visual Studio 2022 is ready to manage version control with its built-in Git features. With this setup, you can confidently create repositories, commit changes, and connect to GitHub whenever you want to publish and share your code with students in a clean and professional way.

Dot Net Tutorials

About the Author: Pranaya Rout

Pranaya Rout has published more than 3,000 articles in his 11-year career. Pranaya Rout has very good experience with Microsoft Technologies, Including C#, VB, ASP.NET MVC, ASP.NET Web API, EF, EF Core, ADO.NET, LINQ, SQL Server, MYSQL, Oracle, ASP.NET Core, Cloud Computing, Microservices, Design Patterns and still learning new technologies.

Privacy Preferences

We and our partners share information on your use of this website to help improve your experience. For more information, or to opt out click the Do Not Sell My Information button below.

Consent | Do Not Sell My Information

Explore our developer-friendly HTML to PDF API | Printed using PDFCrowd | HTML to PDF